

Cross-Dataset Transferability of CE-GAN Augmented Samples for Robust Network Intrusion Detection

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Abstract—Network intrusion detection systems (NIDS) trained on one network environment frequently underperform when deployed in a different environment due to distributional shift and heterogeneous feature spaces across datasets. We propose a cross-dataset augmentation pipeline centred on a Conditional Encoder GAN (CE-GAN) that generates class-conditional synthetic traffic samples on a labelled source domain and transfers them to a target domain through a *FeatureHarmonizer* that combines semantic feature correspondence with a learned fully-connected projection. To mitigate domain shift during generation, a cross-dataset variant of CE-GAN incorporates Maximum Mean Discrepancy (MMD) regularisation in the generator latent space (Scenario C), compared against direct transfer without alignment (Scenario B) and a within-dataset baseline (Scenario A). Classification is performed by a Nash Equilibrium Ensemble combining Random Forest, Extra Trees, and Gradient Boosting with game-theoretic weighting. We introduce the *Transferability Quality Score* (TQS) to measure whether cross-dataset augmentation yields a net benefit relative to domain-specific gains. Experiments across three public benchmark datasets—NSL-KDD, UNSW-NB15, and CIC-IDS2017—show that CE-GAN augmentation consistently improves within-dataset classification (macro F1 up to 0.8132 on CIC-IDS2017), while cross-dataset transfer utility is highly source–target dependent. MMD alignment confers measurable benefit for the UNSW-NB15→NSL-KDD direction (TQS = +195.89) but degrades performance in other directions, highlighting the importance of domain-pair selection in cross-dataset transfer.

Index Terms—network intrusion detection, generative adversarial network, data augmentation, domain adaptation, transfer learning, Nash equilibrium ensemble, Maximum Mean Discrepancy

I. INTRODUCTION

Network intrusion detection systems are a cornerstone of modern cyber-security infrastructure, yet their real-world effectiveness is hampered by two well-documented challenges: *class imbalance* and *domain shift*. Publicly available benchmark datasets exhibit extreme label skew—benign traffic often accounts for more than 80% of all samples—causing classifiers to ignore rare attack categories that are precisely the ones an operator cares most about. Moreover, network traffic statistics are highly environment-specific: the distribution learned from one campus or enterprise network rarely generalises to another, and the feature vocabularies of different datasets seldom align.

Deep generative models, particularly Generative Adversarial Networks (GANs) [1], offer a principled route to synthetic

minority-class oversampling that respects the underlying data manifold better than heuristic interpolation methods such as SMOTE [2]. Conditional GANs [3] extend this to class-targeted generation, making them directly applicable to multi-class intrusion detection.

Cross-dataset knowledge transfer is an equally active research direction. Transfer learning frameworks that leverage labelled data from a data-rich source domain to bootstrap a data-scarce target domain have demonstrated strong results in image recognition [4] and NLP, but their application to network traffic remains challenging because of the heterogeneous, tabular nature of intrusion detection features.

This paper makes the following contributions:

- 1) **CE-GAN**: a Transformer-based Conditional Encoder GAN that jointly trains a generator and a conditional discriminator over tabular network traffic features, producing high-fidelity class-conditional synthetic samples.
- 2) **CrossDatasetCEGAN**: an extension of CE-GAN that adds an MMD penalty between source and aligned target latent codes, encouraging the generator to produce transferable representations.
- 3) **FeatureHarmonizer**: a two-stage mapping module combining expert-defined semantic correspondences with a learned fully-connected projector to bridge heterogeneous feature spaces.
- 4) **Nash Ensemble Classifier**: a three-estimator ensemble (RF, Extra Trees, GBT) whose mixing weights are derived from a Nash Equilibrium, providing principled uncertainty calibration.
- 5) **Transferability Quality Score (TQS)**: a scalar metric that quantifies whether cross-dataset augmentation yields a net performance gain relative to the source-domain improvement already achieved by the GAN.
- 6) An empirical evaluation across three scenarios (within-dataset, direct transfer, and MMD-aligned transfer) on NSL-KDD, UNSW-NB15, and CIC-IDS2017.

II. RELATED WORK

A. GAN-based Data Augmentation for NIDS

Several studies have applied GANs to synthesise minority-class network traffic. Lin *et al.* [5] proposed IDSGAN, which

uses a WGAN architecture to generate adversarial attack traffic for testing IDS systems. CTGAN [6] introduced a conditional tabular GAN that handles discrete and continuous columns, and has been applied to NIDS datasets with measurable improvements in minority-class recall. Our CE-GAN uses a Transformer encoder backbone, enabling it to model long-range dependencies among correlated traffic statistics that fully-connected architectures may miss.

B. Domain Adaptation for Intrusion Detection

Domain adaptation techniques aim to align the feature distributions of source and target domains. Maximum Mean Discrepancy is a non-parametric distribution distance widely used in shallow transfer learning [7] and subsequently adapted for deep networks [8]. For network traffic, Zhao *et al.* [9] applied domain-adversarial training to adapt a NIDS from one network topology to another. Our MMD-regularised Cross-DatasetCEGAN follows a similar intuition but applies domain alignment at the generator latent space rather than the classifier layer, preserving class-conditional structure in the generated samples.

C. Ensemble Methods in NIDS

Ensemble classifiers consistently outperform single models on imbalanced intrusion detection benchmarks [10]. Nash Equilibrium-based weight optimisation in ensembles was introduced by [11] and provides a game-theoretically motivated alternative to simple voting or stacking. Our Nash Ensemble Classifier inherits this weighting mechanism and is the downstream classifier in all three experimental scenarios.

III. METHODOLOGY

A. CE-GAN Architecture

CE-GAN is a conditional GAN designed for tabular traffic features. Let $\mathbf{x} \in \mathbb{R}^d$ denote a feature vector and $c \in \{0, \dots, K-1\}$ its class label.

Generator. Given a noise vector $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_p)$ and a class embedding $\mathbf{e}_c$, the generator maps $G : (\mathbf{z}, \mathbf{e}_c) \mapsto \hat{\mathbf{x}} \in \mathbb{R}^d$. The backbone is a L -layer Transformer encoder with model dimension D , followed by a linear output head. The class embedding $\mathbf{e}_c$ is injected via cross-attention at every Transformer layer, conditioning all generated features jointly.

Discriminator. The discriminator $D(\mathbf{x})$ outputs two heads: a real/fake probability and a K -class softmax. The generator is trained with a cross-entropy auxiliary classification loss in addition to the standard adversarial objective, following the AC-GAN design [12]:

$$\mathcal{L}_G = \mathcal{L}_{\text{adv}} + \lambda \mathcal{L}_{\text{cls}}, \quad (1)$$

where λ balances the two terms.

B. CrossDatasetCEGAN with MMD Regularisation

When a target dataset $\mathcal{T}$ is available without labels, we extend CE-GAN with an MMD penalty between the source latent codes $\{\mathbf{h}_i^{(s)}\}$ and target latent codes $\{\mathbf{h}_j^{(t)}\}$:

$$\mathcal{L}_{\text{MMD}} = \left\| \frac{1}{n_s} \sum_i \phi(\mathbf{h}_i^{(s)}) - \frac{1}{n_t} \sum_j \phi(\mathbf{h}_j^{(t)}) \right\|_{\mathcal{H}}^2, \quad (2)$$

where ϕ is the feature map induced by an RBF kernel. The full generator loss becomes:

$$\mathcal{L}_G^{\text{cross}} = \mathcal{L}_{\text{adv}} + \lambda \mathcal{L}_{\text{cls}} + \gamma \mathcal{L}_{\text{MMD}}, \quad (3)$$

with the MMD weight $\gamma = 0.05$ in all reported experiments. Target samples are aligned to the source feature space by zero-padding or truncation before being passed to the discriminator encoder, as the MMD alignment occurs in the latent space rather than the input space.

C. FeatureHarmonizer

Generated samples live in the source feature space $\mathbb{R}^{d_s}$ but must be injected into a target training set in $\mathbb{R}^{d_t}$. The FeatureHarmonizer resolves this mismatch in two steps.

Step 1: Semantic mapping. An expert-curated correspondence table maps source features to semantically equivalent target features (e.g., `duration` $\rightarrow$ `dur`, `src_bytes` $\rightarrow$ `sbytes`). For the NSL-KDD $\rightarrow$ UNSW-NB15 direction, 23 of the 43 UNSW-NB15 features are covered by semantic mappings.

Step 2: Projected mapping. For target features without a semantic counterpart, a shallow fully-connected network $f_\theta : \mathbb{R}^{d_s} \rightarrow \mathbb{R}^{d_t^{\text{proj}}}$ is trained to minimise reconstruction loss on real target samples, producing the remaining 17 features. Together, semantic + projected mappings cover all 40 non-trivial UNSW-NB15 features, with 3 residual features filled by the FC output.

D. Nash Ensemble Classifier

The Nash Ensemble Classifier combines three base estimators—a Random Forest (RF), Extra Trees (ET), and Gradient Boosted Trees (GBT)—into a mixed strategy by solving a two-player zero-sum game between the ensemble and an adversary that maximises classification error. The optimal Nash mixed strategy assigns weight w_k to estimator k such that no unilateral deviation improves expected accuracy. Weights are initialised performance-proportionally for within-dataset experiments (Scenario A) and uniformly for cross-dataset experiments (Scenarios B and C), where no domain-specific prior is available.

E. Transferability Quality Score

We introduce the TQS to measure whether cross-dataset augmentation provides a net improvement relative to the source-domain augmentation quality:

$$\text{TQS} = \frac{F1_{\text{tgt, aug}} - F1_{\text{tgt, base}}}{F1_{\text{src, within}} - F1_{\text{src, base}}} \times 100, \quad (4)$$

where $F1_{tgt, aug}$ is the target macro F1 with synthetic augmentation, $F1_{tgt, base}$ is the same without augmentation, $F1_{src, within}$ is the source macro F1 with CE-GAN augmentation, and $F1_{src, base}$ is without. A $TQS > 0$ indicates that augmented transfer is beneficial; $TQS < 0$ signals negative transfer where cross-dataset synthetic samples degrade target performance.

F. Three-Scenario Evaluation Protocol

- **Scenario A (within-dataset):** CE-GAN trains and generates on the same dataset. Nash Ensemble is trained on the augmented set and evaluated on the held-out test split.
- **Scenario B (direct transfer):** CE-GAN trains on source; synthetic source samples are passed through FeatureHarmonizer and injected into the target training set. No MMD alignment.
- **Scenario C (MMD-aligned transfer):** CrossDatasetCE-GAN trains on source with target samples as MMD anchors. Otherwise identical to Scenario B.

IV. EXPERIMENTAL SETUP

A. Datasets

Experiments use three widely cited NIDS benchmarks (Table I).

TABLE I
DATASET STATISTICS AFTER PREPROCESSING

Dataset	Features	Classes	Samples (train)
NSL-KDD	41	26	40,000
UNSW-NB15	43	9	40,000
CIC-IDS2017	78	12	40,000

NSL-KDD [13] is a refined version of the KDD Cup 1999 dataset, eliminating duplicate records and providing a more balanced class distribution across 41 TCP/IP connection features.

UNSW-NB15 [14] was generated in a controlled testbed environment and contains 43 mixed-type features covering flow-level, content, and time statistics across nine attack categories.

CIC-IDS2017 [15] was captured in a realistic emulated enterprise network and encompasses 78 flow-level features and twelve traffic classes including DDoS, port-scan, brute-force, and web attacks.

Preprocessing followed a consistent pipeline: (1) removal of infinite and NaN values with column-wise median imputation; (2) rare-class filtering retaining only classes with at least six samples; (3) stratified 80/20 train-test split; (4) StandardScaler normalisation fitted on training data. All datasets were capped at 50,000 samples for computational tractability.

B. Hyperparameters

GAN training: 50 epochs, batch size 256, Adam optimiser with $\beta_1 = 0.5$, learning rate 10^{-4} . Transformer backbone: $D = 128$, $L = 3$ layers. MMD weight: $\gamma = 0.05$. Augmentation: $n_{aug} = 500$ synthetic samples per class. FeatureHarmonizer projector: 200 training epochs. Nash Ensemble: 100 trees per estimator. All experiments were run on a single NVIDIA GPU (CUDA 12.4).

V. RESULTS

A. Within-Dataset Baseline (Scenario A)

Table II (top block) shows that CE-GAN augmentation achieves high accuracy across all three datasets, with macro F1 ranging from 0.5227 on the class-imbalanced UNSW-NB15 to 0.8132 on CIC-IDS2017. NSL-KDD achieves accuracy of 0.9899 and macro F1 of 0.7723, confirming that the Transformer-based generator successfully captures the multi-modal traffic distribution. The lower macro F1 on UNSW-NB15 is attributable to the highly skewed class distribution, where the Normal class constitutes over 87% of the training data; per-class F1 on minority attack categories degrades accordingly.

B. Cross-Dataset Transfer Without Alignment (Scenario B)

Direct transfer via FeatureHarmonizer (Scenario B) preserves classification accuracy within 1–2 percentage points of the corresponding Scenario A baseline in most cases (Table II, middle block). The NSL-KDD→CIC-IDS2017 direction achieves a macro F1 of 0.8144, nearly matching the within-dataset CIC-IDS2017 baseline (0.8132), demonstrating that semantic feature alignment can bridge datasets with 37 additional features. In contrast, the NSL-KDD→UNSW-NB15 direction only reaches $F1 = 0.5222$, reflecting that the generated minority-class samples from NSL-KDD do not improve UNSW-NB15’s inherent class imbalance problem, since the label spaces partially mis-align.

C. MMD-Aligned Transfer (Scenario C)

Comparing Scenarios B and C in Table II, MMD alignment produces mixed results depending on the source–target direction. For UNSW-NB15→NSL-KDD, macro F1 improves from 0.7701 (B) to 0.7897 (C), a gain of +2.0 points. Similarly, UNSW-NB15→CIC-IDS2017 improves from 0.8270 to 0.8313 (+0.4 points). Conversely, NSL-KDD→UNSW-NB15 slightly decreases (0.5222 → 0.5208), and NSL-KDD→CIC-IDS2017 marginally improves (0.8144 → 0.8182). The asymmetry suggests that UNSW-NB15’s richer 43-dimensional feature space provides more effective MMD anchors than NSL-KDD’s 41-dimensional space when bridging to the disparate CIC and NSL distributions.

D. Transferability Quality Score

Table III reports TQS values for Scenario C across evaluated source–target pairs.

The strongly positive TQS for UNSW-NB15→NSL-KDD (+195.89) confirms that MMD-aligned transfer is highly effective in this direction: the target improvement is nearly twice the source-domain gain. This result is aligned with the quantitative improvement seen in Table II and suggests that the richer UNSW-NB15 latent space, when regularised with MMD, successfully captures NSL-KDD-relevant structure.

The strongly negative TQS for UNSW-NB15→CIC-IDS2017 (−476.74), despite the macro F1 improvement in Table II, indicates a methodological artifact: the source-domain augmentation gain ($F1_{src, within} - F1_{src, base}$) was very small (near

TABLE II
MAIN RESULTS: ACCURACY, MACRO F1, AND WEIGHTED F1 ACROSS ALL SCENARIOS

Scenario	Source	Target	Direction	Accuracy	Macro F1	Weighted F1
A (within)	NSL-KDD	NSL-KDD	—	0.9899	0.7723	0.9891
	UNSW-NB15	UNSW-NB15	—	0.9780	0.5227	0.9771
	CIC-IDS2017	CIC-IDS2017	—	0.9965	0.8132	0.9963
B (direct transfer)	NSL-KDD	UNSW-NB15	KDD→UNSW	0.9784	0.5222	0.9772
	NSL-KDD	CIC-IDS2017	KDD→CIC	0.9966	0.8144	0.9964
	UNSW-NB15	NSL-KDD	UNSW→KDD	0.9893	0.7701	0.9886
	UNSW-NB15	CIC-IDS2017	UNSW→CIC	0.9964	0.8270	0.9962
C (MMD-aligned)	NSL-KDD	UNSW-NB15	KDD→UNSW	0.9785	0.5208	0.9771
	NSL-KDD	CIC-IDS2017	KDD→CIC	0.9966	0.8182	0.9964
	UNSW-NB15	NSL-KDD	UNSW→KDD	0.9893	0.7897	0.9886
	UNSW-NB15	CIC-IDS2017	UNSW→CIC	0.9965	0.8313	0.9963

TABLE III
TRANSFERABILITY QUALITY SCORE (SCENARIO C)

Source	Target	TQS	Interpretation
NSL-KDD	UNSW-NB15	−17.72	Slight negative transfer
UNSW-NB15	NSL-KDD	+195.89	Strongly beneficial
NSL-KDD	CIC-IDS2017	−11.05	Marginal negative transfer
UNSW-NB15	CIC-IDS2017	−476.74	Strongly negative

zero), amplifying the denominator sensitivity in Eq. (4). This underscores that TQS is most interpretable when the source domain exhibits a measurable augmentation gain.

E. Comparison with SMOTE Baseline

TABLE IV
CE-GAN VS. SMOTE MACRO F1 ON SELECTED PAIRS

Source	Target	SMOTE F1	CE-GAN B F1
NSL-KDD	CIC-IDS2017	0.6507	0.8144
UNSW-NB15	CIC-IDS2017	0.6507	0.8270

Against the SMOTE cross-dataset baseline (Table IV), CE-GAN augmentation yields substantial macro F1 improvements of +16 and +18 points respectively on the CIC-IDS2017 target, validating the advantage of manifold-aware generative oversampling over linear interpolation.

VI. DISCUSSION

A. When Does Cross-Dataset Transfer Help?

Our results reveal that the utility of cross-dataset CE-GAN augmentation is highly dependent on the semantic relatedness of the source and target feature spaces. The NSL-KDD→UNSW-NB15 direction is the most challenging: both datasets model similar attack types (DoS, Probe, R2L, U2R vs. Reconnaissance, DoS, Exploits) but differ in traffic capture methodology, feature semantics, and class distribution. The FeatureHarmonizer covers all 43 UNSW-NB15 features, yet the macro F1 (≈ 0.52) does not exceed the no-augmentation baseline, suggesting that even correctly-aligned features carry

domain-specific statistical artefacts that the Nash Ensemble cannot ignore.

In contrast, CIC-IDS2017 benefits from both NSL-KDD and UNSW-NB15 augmentation. CIC-IDS2017’s 78-feature space has low semantic overlap with either source, yet the FC-projected mapping generalises well, possibly because CIC-IDS2017 contains a larger number of attack classes and thus benefits more from any additional minority-class coverage.

B. Role of MMD Alignment

MMD regularisation consistently benefits the UNSW-NB15 source but not the NSL-KDD source. One explanation is that NSL-KDD’s pre-2000 synthetic attack traffic is distributionally very different from both modern benchmark targets, so MMD alignment in a common latent space introduces a misaligned signal that corrupts the generator. UNSW-NB15’s controlled testbed captures are closer in distribution to both NSL-KDD (which is itself based on a similar simulated environment) and CIC-IDS2017, making MMD anchoring effective.

C. Limitations

Several limitations bound the conclusions of this work. First, label spaces across datasets only partially overlap, and our current label remapping (clamp to target class range) is a coarse approximation. A cross-dataset ontology mapping [16] could substantially improve label alignment. Second, TQS denominator sensitivity makes it unreliable when source-domain augmentation provides negligible gain, as observed in the UNSW-NB15→CIC-IDS2017 pair. Third, all experiments used 50 training epochs and fixed architecture hyperparameters; longer training or architecture search might close the gap for the harder transfer directions.

VII. CONCLUSION

We presented CE-GAN, a Transformer-based conditional GAN for class-conditional network traffic synthesis, and CrossDatasetCEGAN, its MMD-regularised extension for cross-dataset transfer. Together with FeatureHarmonizer feature alignment and Nash Ensemble classification, the pipeline

achieves within-dataset macro F1 of up to 0.8132 and demonstrates that cross-dataset augmentation can be beneficial—macro F1 improving by +2.0 points for UNSW-NB15→NSL-KDD—when feature spaces are semantically related and MMD anchors are reliable.

The introduced TQS provides an interpretable scalar verdict on transfer quality, though its formulation requires a non-negligible source-domain augmentation gain to remain numerically stable. Future work will explore attack-category-aware label mapping across datasets, conditional MMD alignment, and adversarial domain discriminators to further close the cross-dataset performance gap.

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